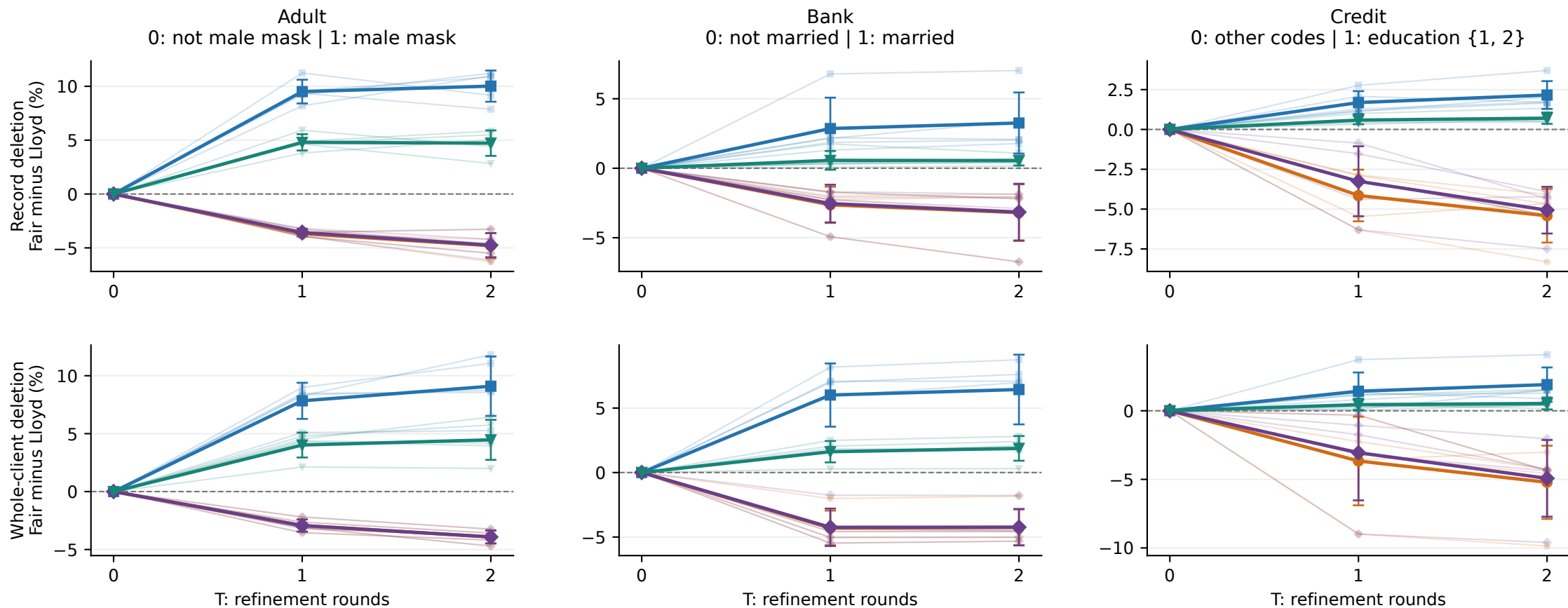


Guarded fair versus ordinary Lloyd updates

Identical fair-initialized centers and retained records; $k = 10$; five paired seeds per dataset and population.

—●— Group 0 cost —■— Group 1 cost —◆— Worst-group cost —▼— Population-average cost



Each point uses $100 \times (\text{fair cost} - \text{Lloyd cost}) / \text{Lloyd cost}$ within seed. Negative means lower cost with fair updates.

Thin paths: all five seeds. Bold markers: mean paired percentage difference. Bars: one sample SD; dataset scales differ.

T0 is the shared initialization. The studies have distinct survivors. Absolute costs, exact fractions and every case are in the tables.

Fair updates lower worst-group cost here, with higher population-average and group 1 costs. This tests update choice, not initialization.